

Introduction

This analysis uses product data from an online sports apparel retailer. The information describes different aspects of the products sold and is provided in four separate datasets: brands, finance, info, and reviews. Each dataset contains 3,179 observations and includes a `product_id` variable.

Study Objectives

1. Examine the number of products and average revenue for Adidas and Nike across listing price quartiles.
2. Assess whether differences exist between product description length and average product rating.
3. Build and compare three predictive models to determine which model best predicts revenue.

Data Understanding and Preparation

First, the four datasets were merged using an inner join on `product_id`. After merging, the dataset contained 3,179 rows, confirming that the tables were combined correctly.

Next, 70 duplicate rows were identified and removed. Missing values were then examined. Because one observation contained a missing value in the target variable (revenue), that row was removed. After this step, no missing values remained in the dataset.

The target variable (revenue) represents the total income generated by a product and is calculated as units sold multiplied by the sale price.

The distribution of revenue is highly right-skewed. To stabilize the distribution and improve regression modeling, the variable was transformed using $\log(\text{revenue} + 1)$. The constant 1 was added to allow the transformation to be applied to observations with zero revenue.

The dataset includes three product-level variables: `product_name`, `description`, and `brand`. The `product_name` variable identifies the item. The number of unique product names is smaller than the total number of observations, indicating that some products may appear more than once in the dataset. The `brand` variable is categorical and includes two categories: Adidas (83%) and Nike (17%).

The `description` variable contains unstructured text describing the product. To incorporate this information into the predictive models, the number of words in each description was calculated and stored in a new variable, `description_length`.

Three variables describe product pricing: `listing_price`, `discount`, and `sale_price`. All three variables exhibit right-skewed distributions.

Data screening identified 343 observations with `listing_price` equal to zero. Because a zero listing price is not economically meaningful, these values were treated as missing. The missing values were then imputed using `sale_price` and `discount` (see formula in the Appendix). After this step, no missing values remained in the dataset.

Among the price-related variables in the dataset, `sale_price` is less suitable for prediction because it represents the final transaction price and is directly used in the calculation of revenue. For this reason, `listing_price` and `discount` provide more meaningful predictors, as they reflect the retailer's initial pricing decision and subsequent price adjustments. Therefore, `sale_price` was excluded from the predictive models.

The dataset also includes the variables rating and reviews. The rating variable represents the average customer rating on a five-point scale ranging from 0 to 5, while reviews records the number of customer reviews for each observation.

To better understand the meaning of the reviews variable, the relationship between reviews and an estimated number of units sold (`units_sold`) was examined (Figure 1). The `units_sold` variable was estimated using the formula provided in the Appendix. The results show a perfect linear relationship ($R^2 = 1$), with reviews approximately equal to $0.55 \times \text{units_sold}$. This indicates that the number of reviews closely reflects the number of products sold. Because this variable acts as a proxy for `units_sold`, which is a component of revenue, including reviews as a predictor would introduce data leakage. Therefore, this variable was excluded from the predictive models.

The final set of predictors includes one categorical (brand) and four continuous variables: `description_length`, `listing_price`, `discount`, and `rating`. The scatterplot and correlation matrix show that correlations among the continuous predictors are generally weak, indicating that multicollinearity is not a concern for the subsequent predictive modeling (Figure 2).

To further explore potential relationships between the predictors and the target variable, distributions of the variables were examined interactively (Figure 3). Visual inspection highlights several patterns. Observations with the lowest revenue appear more frequently among Nike products and are often associated with missing ratings (Figure 3A). Lower listing prices occur more frequently among Adidas products, suggesting that Nike items tend to be priced higher (Figure 3B). In addition, Nike products rarely appear with discounts, whereas Adidas products are more often sold with discounts (Figure 3C).

To answer the first case question, products were grouped into listing price quartiles (see formula in the Appendix). The number of products and the average revenue were then compared for Adidas and Nike within each quartile (Figure 4). For Adidas, average revenue increases steadily across listing price quartiles, with the highest values observed in the fourth quartile ($p < 0.001$). For Nike, a similar pattern is observed, although the relationship appears less stable due to the smaller number of Nike products in the dataset. Additionally, the product distribution across quartiles differs between brands: Adidas offers a relatively balanced assortment across price segments, whereas Nike products are more concentrated in higher price quartiles (Table 1). Sales volumes also display different patterns across the two brands. For Adidas, the highest number of units sold occurs in the lower price quartiles, with volumes declining as prices increase. In contrast, Nike shows a gradual increase in units sold toward the higher price quartiles, suggesting relatively stronger demand for higher-priced products.

To examine whether product description length is associated with mean rating, products were grouped into description length quartiles (see formula in the Appendix). The results show no statistically significant differences in mean ratings across description length quartiles ($p = 0.2602$ for Adidas and $p = 0.8370$ for Nike) (Figure 5).

Modeling and Evaluation

Predicting Revenue (Regression Models)

A validation column was created to partition the data into training (65%), validation (20%), and test (15%) sets using a random split with random seed 12345 (Figure 6). Three models were selected to represent different predictive modeling approaches: linear, tree-based, and non-linear methods. Model performance across the three approaches is summarized in Figure 7.

The first model estimated was a linear regression model including the selected predictors and interaction terms (Figure 8). The results indicate that brand, rating, listing price (listing_price_imp), and discount are statistically significant predictors of revenue, along with interactions between brand and rating and between brand and listing price. The model demonstrates good predictive performance, with $R^2 = 0.667$ for the validation set and 0.640 for the test set, and the overall model is statistically significant ($p < 0.0001$). The standardized coefficients further suggest that brand and customer ratings are the most influential factors associated with product revenue.

The second model estimated was a regression tree model predicting $\log(\text{revenue}+1)$ using the selected predictors (Figure 9). The final tree with seven splits was selected based on validation and test set performance. The model achieved an R^2 of 0.663 on the validation set and 0.680 on the test set, indicating predictive performance comparable to the linear regression model. The tree structure highlights rating and brand as key predictors of revenue. Variable importance analysis further indicates that brand and rating contribute the largest reductions in prediction error, while listing price and discount contribute less to reducing prediction error.

The third model estimated was a feed-forward neural network with one hidden layer containing five neurons and a *tanh* activation function (Figure 10). Continuous predictors were transformed using Johnson transformations to approximate normality prior to training the neural network. The model was trained using 10 tours. The neural network achieved an R^2 of 0.706 on the validation set and 0.721 on the test set, indicating the strongest predictive performance among the models considered. To interpret the neural network, SHAP (Shapley Additive Explanations) values were computed to assess the contribution of each predictor to the model predictions. The results indicate that brand has the strongest contribution to predicted revenue, followed by rating and listing price, while discount and description length have smaller effects.

In addition to identifying important predictors of revenue, the models also revealed interactions between the variables. The strongest interaction occurs between brand and rating. For Nike products, higher ratings are associated with higher predicted revenue. For Adidas products, predicted revenue remains relatively high across a broader range of ratings, suggesting that brand strength plays a larger role in revenue performance.

A second interaction appears between listing price and rating. Products with higher ratings are able to sustain higher listing prices while maintaining stronger predicted revenue. In addition, the interaction between brand and listing price suggests that price increases are associated with higher predicted revenue for Adidas products, while the relationship appears weaker for Nike products.

Although all models identify similar interaction patterns, they represent these relationships in different ways (Figure 11). The linear regression model shows interactions in a linear form, the regression tree represents them as stepwise changes resulting from split rules, and the neural network captures smoother nonlinear patterns in how the variables jointly influence predicted revenue.

Overall, the results indicate that all three models provide strong predictive performance for estimating product revenue. The neural network achieved the highest predictive accuracy on both the validation and test sets, suggesting that it is best able to capture complex relationships in the data. However, the linear regression and regression tree models produced comparable results while offering greater interpretability. Across all models, brand, rating, and listing price consistently emerged as the most important predictors of revenue.

The analysis also shows that these variables interact in meaningful ways: the effect of rating differs across brands, and higher-rated products are better able to sustain higher listing prices. These findings suggest that both brand strength and customer perception play important roles in revenue performance and should be considered when evaluating pricing and product positioning strategies.

Low-Revenue Risk Prediction (Classification Models)

To complement the revenue prediction analysis, additional models were developed to identify products at risk of generating low revenue. A binary variable (`low_revenue_flag`) was created, where products with revenue at or below the 25th percentile were labeled as 1 (see formula in the Appendix). A validation column was created to partition the data into training (65%), validation (20%), and test (15%) sets using a stratified split based on the `low_revenue_flag` variable to preserve the class distribution (random seed 12345) (Figure 12).

Using the same predictors as in the regression models, three classification models were developed: logistic regression, a decision tree, and a neural network. In this case, the target variable was the binary `low_revenue_flag`. Model performance was evaluated using several classification metrics. The results indicate that all three models demonstrate similar predictive ability, with AUC values ranging from approximately 0.81 to 0.83 (Figure 13). Among the models, the neural network shows slightly stronger overall performance (Entropy RSquare = 0.3329, AUC = 0.8319), while the decision tree and logistic regression exhibit comparable discrimination between low-revenue and non-low-revenue products. However, practical use of these models requires selecting a probability threshold to classify products as low-revenue or not.

Because the objective of this analysis is to identify products at risk of very low revenue, sensitivity becomes particularly important. At the same time, marketing interventions aimed at improving sales may involve additional costs, making precision also relevant. Therefore, different classification thresholds were evaluated to determine a balance between sensitivity and precision (Table 2). The comparison shows that a classification threshold of 0.35 provides the most favorable balance between sensitivity and precision. At this threshold, the neural network achieves one of the highest sensitivity values (0.6163), equal to the decision tree, while also demonstrating the highest precision (0.7067) and the highest F1 score (0.6584) among the evaluated thresholds (Figure 14).

To complement the neural network results, the decision tree model was additionally considered as an interpretable tool that can help explain the characteristics of products at risk of very low revenue (Figure 15). The tree indicates that brand is the primary factor separating product segments. For Nike products, the risk of very low revenue is mainly associated with low product ratings, particularly below approximately 2.6. In contrast, for Adidas products, the probability of very low revenue is more strongly related to listing price levels, with additional segmentation driven by discount levels. In particular, products with relatively low prices combined with limited discount adjustments appear more frequently in the lowest revenue quartile. This suggests that insufficient discounting for certain price ranges may reduce product competitiveness, increasing the likelihood of very low revenue outcomes.

Conclusion

Overall, the analysis suggests that revenue performance is primarily driven by brand positioning, product ratings, and pricing strategy. The results also indicate that longer product descriptions do not improve customer ratings, suggesting limited value from expanding product text alone. From a business perspective, monitoring product ratings and adjusting pricing or discount strategies for underperforming products may help reduce the risk of very low revenue.